

Real national income (Y) and the price level

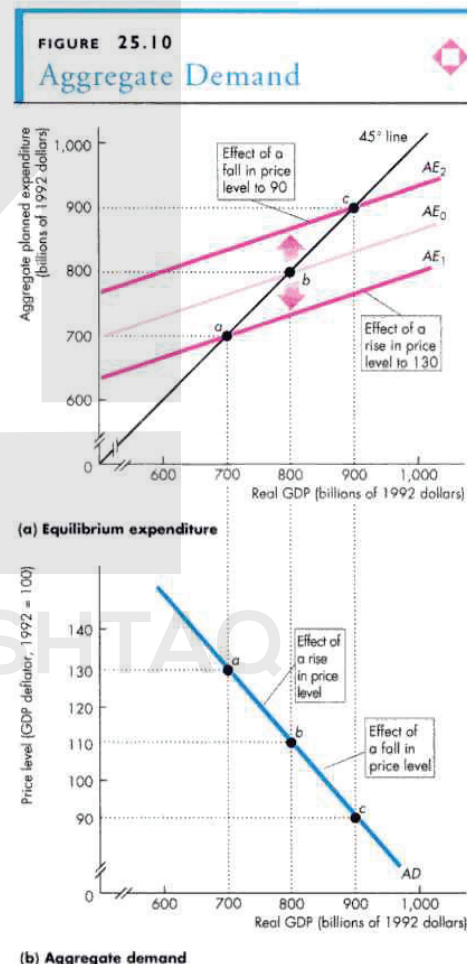
If we drop the assumption of constant price level and allow the prices to change then the aggregate expenditures will be influenced as the price level changes. A rise in the price level shifts the AE curve downwards, while a fall in the price level shifts it upwards. In other words price level and the desired AE are negatively related to each other.

The first reason why changes in the price level affect the AE is the wealth effect of changes in the price level. A higher price level reduces the real value or purchasing power of public's accumulated financial assets. In particular the real value of assets with fixed money values, such as saving account or bonds, reduce. Because of the fall in the real value of such assets, the public is poorer in real terms and will cut back on its spending i.e. the consumption expenditures will fall. Conversely, a decline in the price level will increase the real value or purchasing power of a person's wealth and increase spending.

Secondly, as the price level rises, so do interest rate, and rising interest rates reduce certain kinds of consumption and investment spending. When the price level increases consumers need more money for purchases. With a fixed supply of money, this increase in demand for money drives up the interest rate. Higher interest rates curtail interest-sensitive expenditures by businesses and households e.g. consumers will decide not to purchase houses or automobile because of the rise in interest rate.

The volume of imports and exports depend on relative price levels between domestic country and abroad. If the price level rises in the UK relative to foreign countries, UK buyers will purchase more imports and fewer UK goods. Similarly the rest of the world will buy fewer UK goods, reducing UK exports. In brief, a rise in our price level will increase our imports and reduces our exports, reducing the amount of net exports. This will result in the decline in the aggregate amount of domestic output demanded.

So it can be concluded that the price level and the real national income (Y) are negatively related i.e. a change in the price level changes equilibrium RNI in opposite direction, holding all other exogenous variable constant.

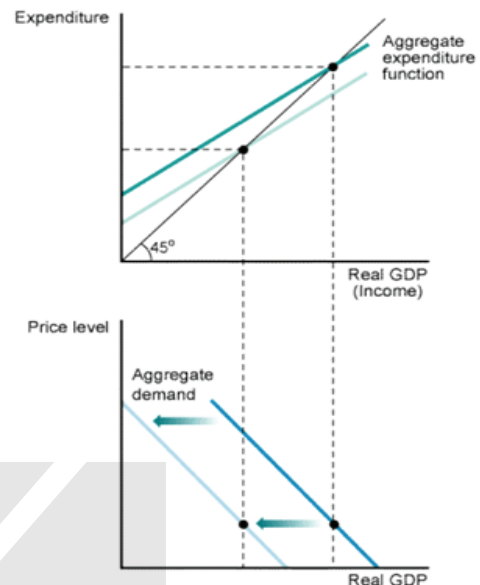


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Shifts in AD curve:

Since the aggregate demand curve plots equilibrium national income as a function of the price level, anything that alters equilibrium national income at any specific price level must shift the aggregate demand curve. In other words any change in autonomous expenditures causes the aggregate expenditure curve to shift and will also cause the aggregate demand curve to shift. A change in autonomous expenditure changes equilibrium national income for any given price level and the simple multiplier measures the resulting horizontal shift in the aggregate demand curve.



Why is the AD curve downward sloping: Remember that for each of these explanations, we are talking about a movement along the aggregate demand curve, not a shift in the curve (see the next section). The initial move is always a change in the price, which then causes the following three things to happen.

1. *Interest rates:*

If the price level rises, the rate of interest rises in an effort to stop the price rise getting out of hand. Consumers' mortgage payments will rise (in the UK, the majority of people have variable rate mortgages). This means that consumers will have less money left over to buy goods and services. Higher interest rates make it more expensive for consumers to buy 'big ticket' items, like new cars, for which money tends to be borrowed. In the same way, firms are less likely to borrow for investment. So, in response to a rise in the price level, the final effect is a fall in planned consumption and investment, and so a reduction in AD.

2. *Changes in real wealth:*

If the price level rises, for a given level of nominal wealth, then peoples' real wealth will fall. This will be true for incomes too if nominal incomes remain constant. Basically, people will feel less well off, and so will probably consume fewer goods and services.

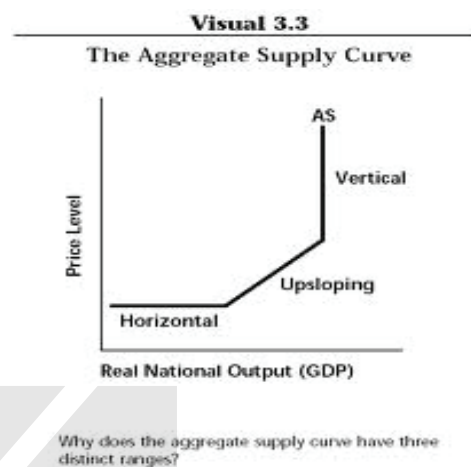
3. *The foreign sector:*

For a given exchange rate, if the UK price level rises, then home produced goods will become relatively more expensive in other countries, and so the demand for exports will fall. Also, imports into the UK from other countries will appear relatively cheaper; because their price level has remained unchanged while the price of the home produced goods are rising. The demand for foreign imports will, therefore, rise. Both of these effects cause the level of real national income to fall as the price level rises.

There is no such thing as luck, believe in hard work

Real national income (Y) and the price level

Aggregate supply: It is a schedule, graphically represented by a curve, showing the level of real domestic output which will be produced at each price level. Short-run aggregate supply curve is drawn on the assumption that the prices of all the factors of production that firms use, such as labor, remain constant. Higher price levels create incentive for firms to produce and sell additional output, while the lower price levels reduce output. As a result, the relationship between the price level and the amount of real output firms offer for sale is direct or positive. The three segments of the AS curve are identified as (1) the horizontal, (2) up-sloping (the intermediate), and (3) the vertical range.



The horizontal range of AS comprises real levels of output substantially less than the full employment level of output i.e. the vertical line. Thus, the horizontal range implies the economy is in a severe recession and that large amounts of unused machinery and equipments and unemployed workers are available for production. These idle resources can be put back to work with little or no upward pressure on the price level.

The up-sloping range, an expansion in real national income is accompanied by a rising price level. As the economy expands the firms may encounter shortages of skilled labors, or raw materials or other production bottleneck may begin to appear. Additional workers will create congestion in the workplaces and productivity will decline, increasing per unit cost and product prices.

Finally, the economy reaches full employment level of output at vertical range. Any increase in price level in this range will fail to increase real output because the economy is operating at full capacity. Individual firms may try to expand production by bidding up resources away from other firms. This increase in resource prices (costs) will lead to an increase in price level but the real output will remain unchanged.

Real national income and the price level in the long run:

In the long-run when all the resources are employed at their normal rates of utilization, the economy produces its potential output or full employment level of output. When the actual output diverges from the potential output, the gap is called the GDP gap or output gap.

The deflationary gap: If the equilibrium level of income is less than the full employment level of output, there will be excess capacity in the economy and hence demand unemployment. This will reduce the prices of the factors of production and their demand is low. At potential level of national income the gap by which the aggregate expenditure fall short of the national income is called the deflationary gap.

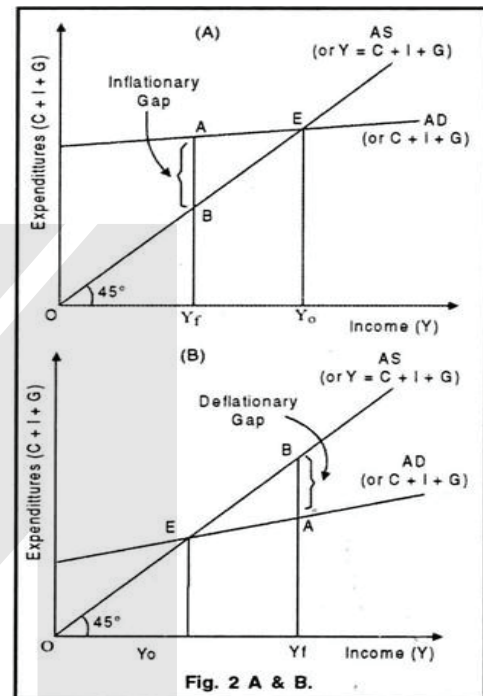
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Real national income (Y) and the price level

If there is deflationary gap there is recession in the economy causing the prices of inputs to fall and in the economy causing the prices of inputs to fall and if the economy is left on its own for recovery than the aggregate supply adjusts itself in response to the fall in input prices. But this process of recovery is sluggish and it may cause recession to prolong.

Inflationary gap: An inflationary gap exists when the demand for goods and services exceeds production due to factors such as higher levels of overall employment, increased trade activities, or elevated government expenditure. Against this backdrop, the real GDP can exceed the potential GDP, resulting in an inflationary gap. The inflationary gap is named as such because the relative rise in real GDP causes an economy to increase its consumption, leading prices to climb in the long run.

The inflationary gap represents the point in the business cycle when the economy is expanding. Due to the higher number of funds available within the economy, consumers are more inclined to purchase goods and services. As demand for goods and services increases but production has not yet compensated for the shift, prices rise to restore market equilibrium.



Investment expenditure and interest rate

Investment to an economist is a precise term which involves the acquisition of capital goods designed to provide us with consumer goods and services in the future. Investment spending involves a decision to postpone consumption and to seek to accumulate capital which can raise the productive potential of an economy. The central message of economic theory and the evidence from business surveys is that capital investment is determined by the relationship between the expected returns from investment and the expected costs of financing the investment. In other words, profit maximizing businesses operating in the private sector of the economy will be prepared to go ahead with an investment if they believe that the project will over its projected lifetime yield a real rate of return greater than if the money tied up in an investment project had been invested in the next best alternative way.

For government sector investment, the priorities may be a little different. Public sector investment projects are still subject to test about their expected rates of return, the cost-benefit analysis will normally include estimates of the social costs and benefits of the investment rather than a narrow focus on private costs and benefits.

Investment in economic theory includes three distinct components:

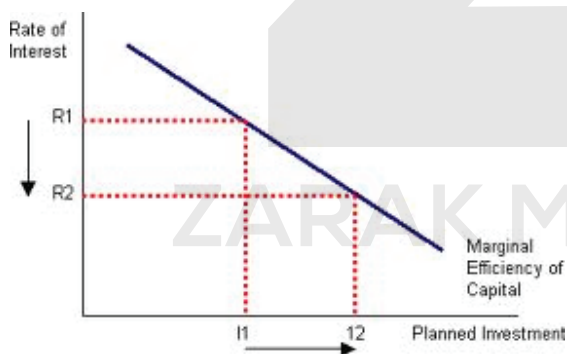
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Real national income (Y) and the price level

1. **Gross domestic fixed capital formation:** All production uses capital goods. These are manufactured aids to production, such as machines, computers, and buildings. The economy's total quantity of capital goods is called the capital stock. Creating new capital goods is an act of investment and is called fixed investment or fixed capital formation which is brought about by positive net investment i.e. investment over and above the depreciation. Like labor, capital also displays the law of diminishing returns.

The return on capital employed is called marginal efficiency of capital (MEC) and is expressed as a percentage of capital. The MEC is defined as the rate of interest which makes a proposed investment viable "at the margin". This is illustrated in the diagram below. At lower interest rate more capital projects appear financially viable because the cost of borrowing money finance the investment is lower and the opportunity cost of using retained profits as an internal source of investment finance is also reduced. A fall in interest rates should (*ceteris paribus*) lead to an expansion along the investment demand curve.

The limited statistical evidence available is that the demand for new capital goods tends to be interest inelastic i.e. there is only a weak link between changes in interest rates and fluctuation in planned capital investment by businesses. Partly this is because many firms prefer to use the capital market through the issue of new shares and bonds to raise funds for investment rather than relying on bank loans. The rate of interest can does affect capital investment decisions, perhaps through its effect on business confidence and also expectations of changing demand and the links between interest rates and the exchange rate. So a period of lower interest rates might stimulate more investment because of expectations of rising consumer demand and a lower exchange rate which will boost export demand.



2. **Investment in stocks:** It includes stocks of raw materials, work in progress and finished goods. Depending on daily supply of raw materials could be expensive or on a given day raw materials do not arrive at the factory. Holding a stock of raw materials and components smoothes the production process. Stocks of finished goods allow firms to meet short run fluctuations in demand. If stocks are not held and if demand exceeds current output, some customers will have to wait and they may imply loss of customers. Investment in stocks represents money tied up as goods. The opportunity cost of a stock is interest sacrificed.

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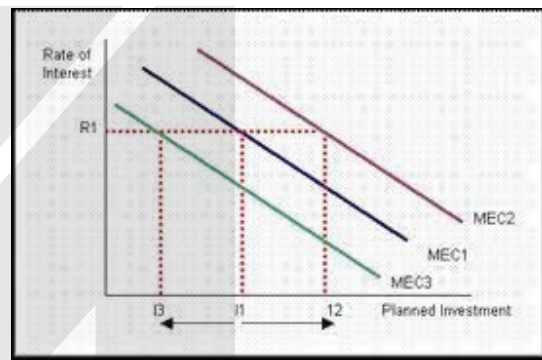
Higher is the interest rate, higher will be the opportunity cost of holding a given stock. So, the investment in stocks is very sensitive to changes in interest rate.

3. **Residential investment:** A house or a flat is a durable asset that yields its utility over a long period of time. This meets the definition of investment, so housing construction is counted as investment expenditure rather than as consumption expenditure. Investment in housing is very sensitive to changes in mortgage rate.

$$\text{Investment} = \text{GDFCF} + \text{stocks} + \text{housing} = f(r)$$

Changes in investment

1. **Price and productivity of capital:** If capital is expensive and labor is relatively cheaper, then business organizations will use labor intensive method of production, as done in many developing countries. This will lead to a leftward shift in the MEI curve. On the other hand, if capital is relatively cheaper, businesses will use capital intensive method of production leads to a rightward shift in MEI curve. The cost will minimize when $MPK/P_k = MPI/PI$



2. **Changes in technology:** When technology changes, using old equipments will have higher unit cost and business organizations will be forced to use equipments with new technology as this will reduce unit cost. This will shift the MEI curve rightwards.
3. **Expectations:** Since investment is made in order to produce output for the future, investment must depend on firm's expectations about future market conditions. If the business is optimistic about future, MEI curve will shift to the right.
4. **Infrastructure:** The pre-requisite for investments in the manufacturing sector is a well established and developed infrastructure. An excellent road network is required to transport raw materials as well as finished goods. For example: Zaire is one of the richest countries in the form of natural resources. Poor infrastructure makes it difficult to exploit such natural resources.
5. **Law and order:** Business organization will always invest, if such investments are secure and profitable. If a country has law and order problem or if there is political instability, the investors will avoid investing in such an economy. Domestic residents will move their investments abroad causing reduction in investments.
6. **Government policy:** Taxes and subsidies greatly affect investments in an economy. A lower tax on the profits of companies leaves a greater after tax profit encouraging more investments. Similarly, subsidies by the government also help in reducing cost causing investments more profitable.

There is no such thing as luck, believe in hard work

Real national income (Y) and the price level

7. **Profits:** The level of business profitability is a major factor driving investment demand. Higher profits suggest a more favorable business climate which boosts business confidence. Higher profits also make it easier for firms to reinvest their 'producer surplus' to fund capital projects.
8. **External economic factors:** The health of the global economy is becoming an important factor influencing capital spending decisions. For example, businesses will consider where to locate their new investments - focusing on factors such as relative costs levels, corporate taxation regimes in different countries, expectations of income and demand growth in different regions and nations and the volatility of exchange rates.

The paradox of thrift: People traditionally think that the more a person saves his income or wealth, the more the amount of income gained in the future. Savings can increase future income as well as output.

The more we save, the more income we shall have!

Based on the Keynesian functions, savings depends on income. We have to cut our consumption expenditure in order to save more. By the multiplier effect, the fall in consumption will lead to a fall in income and with this smaller amount of income, we can then save less than before.

The more we save, the less income we shall have as a result

It is paradoxical to find that: If people try to increase their savings, they will end up saving less. It then follows that: When times are tough (economically unfavorable), we should not tighten our belts. It is called a paradox (of thrift) because the act of a single person to save more for the future is not directly applicable to the whole economy.

Thrift for an individual is fine but not for the economy.

1. What is good for one may not be good for all. A good e.g. is the taking of medicine. What is good for you to take a certain medicine may not be good for others.
2. Saving is good for a person but is not desirable for the whole economy
3. Economic situation
4. The traditional view carries the concept of self-adjustment of the economy with savings and investment moved in the same direction to maintain an equilibrium output.

Whereas the Keynesian view argued that there is no apparent linkage between savings and investment. That is, when savings rises and consumption expenditure falls, the multiplier effect leads to a fall in income also. The amount of investment is not affected by income. As a result the paradox may exist.

The accelerator theory focuses on induced investment and emphasizes the volatility of investment. It states that investment depends on the rate of changes in income (and therefore

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Real national income (Y) and the price level

consumer demand), and that a change in GDP will cause a greater proportionate change in investment. If a \$1 million increase in GDP causes induced investment to rise by \$3 million, the accelerator coefficient is said to be 3.

If GDP is rising, but at a constant rate, induced investment will not change. This is because firms can continue to buy the same number of machines each year to expand capacity. However, a change in the rate of growth of income can have a very significant influence on investment. Table 41.1 provides an example. It is assumed that the firm starts the period with eight machines, that one machine wears out each year and that each machine can produce 100 units of output per year.

Year	Consumer demand	Machines at start of period	Number of machines required	Replacement investment (depreciation)	Induced investment	Total investment
1	800	8	8	1	0	1
2	1000	8	10	1	2	3
3	1600	10	16	1	6	7
4	1800	16	18	1	2	3
5	1800	18	18	1	0	1
6	1700	18	17	0	0	0

Table 41.1: Changes in investment

The table shows that when demand for consumer goods rises by 25% (from 800 to 1000) in the second year, demand for capital goods rises by 200% (from 1 to 3). When the rate of growth in demand for consumer goods slows in year 4, demand for capital goods falls. By year 6, investment falls to zero with the worn-out machine not being replaced, and so productive capacity is reduced. However, an increase in demand for consumer goods does not always result in a greater percentage change in demand for capital goods. For instance, firms will not buy more capital goods if they have spare capacity or if they do not expect the rise in consumer demand to last. It may also not be possible for firms to buy many capital goods if the capital goods industries are working close to capacity. In addition, with advances in technology, the capital-output ratio may change, with fewer machines being needed to produce a given output.

The concept of **multiplier and accelerator** are interrelated i.e. there is an interaction between them. An increase in investments leads to a multiplier time's increase in real national income. This increase in national income will trigger off induced investments causing RNI to increase again. But national income cannot increase at an increasing amount every year because as actual output moves toward potential output the growth in real national income flows down which will lead to a fall in investments and the process will start moving in opposite direction.

The Keynesian liquidity preference theory

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Real national income (Y) and the price level

The demand for money: The demand for money was not related solely to the demand for new capital goods and the supply of funds was not dependent solely on the current level of saving. Money was demanded not simply because it was a medium of exchange, it was also in demand as an asset. The demand for money, is the demand for money to hold. The demand for money refers to a desire to hold wealth in the form of money. The opportunity cost of holding any money balance is the extra interest that could have been earned if the money had been used instead to purchase bonds.

The important motives for holding money balances are:

- **The transactions motive:** Money balances held to pay for regular purchases of goods and services is called transactions balance. The actual money balance held for this purpose obviously depends on households' income. Sum of all households' income is the national income. So the transactions demand is positively associated with national income.

$$T = F(Y)$$

Transactions demand for money also depends on the rate of interest. Higher the rate of interest, lower is the demand for transaction balance demanded because the opportunity cost of holding money will be high.

Transactions demand for money also depends in the level of wealth. Greater the amount of wealth, the more will be the transactions demand for money.

The increased use of credit cards in recent years has reduced the transactions demand for money

Where, Y =Income, r =rate of interest, w =wealth, p =price level.

- **The precautionary motive:** Much expenditure arises unexpectedly, such as when your car breaks down or you have to make an unplanned journey to visit a sick relative. Precautionary balances provide a cushion against uncertainty about the timing of cash flows. If a household does not have extra money balances, then the cost of raising short term money could be difficult or very expensive. To avoid contingencies, most households keep an extra money balance which is precautionary demand for money. The amount demanded depends on the income of households; higher the income of households, the higher will be the precautionary balances.

Of course, if households are lucky, such emergencies may never arise. So, the money balance held remains idle and the opportunity cost of an idle balance is the interest forgone. The precautionary demand for money is positively related with income and negatively related with the rate of interest.

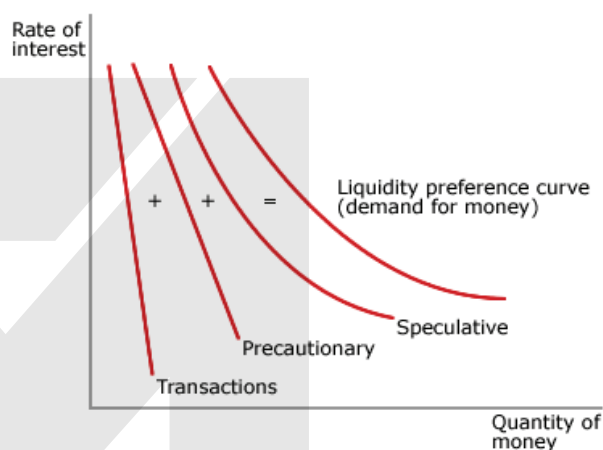
- **The speculative motive:** Over and above transactions balances and precautionary balance a household may have income leftover. It is demand for money as 'store of wealth.' Wealth can be held (stored) in the form of landed property, bonds, etc.

There is no such thing as luck, believe in hard work

Real national income (Y) and the price level

For the sake of simplicity, all forms of assets except money may be clubbed in a single category called bonds. How to make best use of both, considering if we deposit cash in saving bank account, we earn interest and if we purchase bonds, we get monetary return on it. People compare rate of return on bond with rate of interest on bank deposits. It is speculation about future changes (rise/fall) in interest rate and bond prices that the resulting demand for money is called 'speculative demand for money'. Clearly, the aim is to make money (monetary gain) out of money.

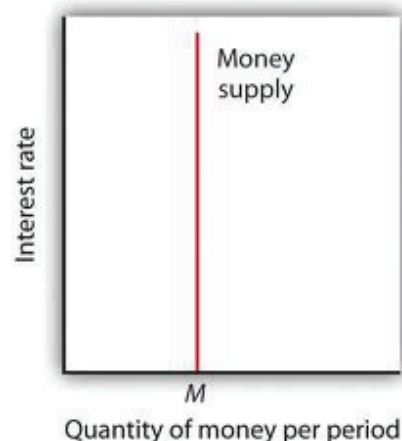
Relationship between bond price and interest rate: Price of a bond is inversely related to market rate of interest. How? Suppose, Rs 1,000 bond yields fixed return of 10% per annum which means the bond has fixed annual Income of Rs 100. Let us assume that the rate of interest in saving bank account falls from Rs 10% to 8%. The question is how much money kept in bank will fetch interest of Rs 100 (equal to return from bond) after one year?



Suppose, X is the amount, then the amount will be $X \times 8/100 = 100$ or $X = \text{Rs}1,250$. Clearly, Rs 1,250 kept in saving bank account will give the same return (i.e., Rs 100) as Rs 1,000 invested in bond. Naturally, people will prefer to buy bond than to deposit cash in bank. Competition among buyers will push up the price of Rs 1,000 bond (face value) to 1,250. Hence, conclusion is obvious.

With fall in the market rate of interest (i.e., bank deposit interest) price of bond rises and vice versa. In other words, price of bond is inversely related to market rate of interest. Mind, speculation about changes (rise/fall) in interest rate and bond prices gives rise to speculative demand for money.

Supply of money: The monetary base (or high powered money) consists of cash notes and coins in circulation outside the central bank. It is sometimes referred to as the narrow monetary base to distinguish it from the wide monetary base, which also includes banks' balances with the central bank. In the UK, the wide monetary base is known as M0. The most usual measure that countries use for money supply is broad



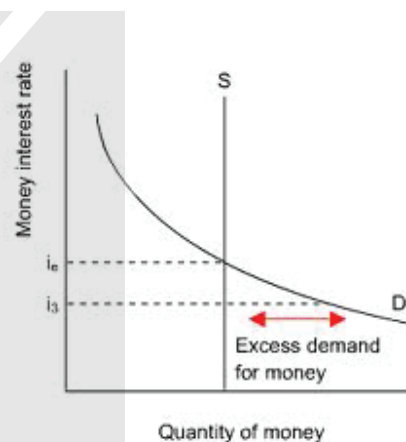
There is no such thing as luck, believe in hard work

Real national income (Y) and the price level

money, which includes both time and sight deposits, retail and wholesale deposits (retail deposits are the loans made by banks at published interest rates and whole sale deposits are the deposits and loans made by and to firms at negotiated interest rates and whole sale deposits are the deposits and loans made by and to firms at negotiated interest rates, and bank and building society saving institutions deposits. In the UK this measure of broad money is known as M4.

The money supply is a stock of pounds per period of time. The monetary authorities control the total money stock directly or in other words, the money stock is exogenously fixed by the policy makers.

The monetary equilibrium: The monetary equilibrium occurs at a point where demand for money equals the supply of money. The rate of interest does the job of equating the quantity of money demanded and the available supply i.e. it does the job of producing monetary equilibrium. The money supply is fixed exogenously at the specific quantity, M_0 because (by assumption) the money supply does not change with the rate of interest; the supply of money is vertical at quantity M_0 . The curve marked as LP is the liquidity preference function which shows how demand to hold money varies with the rate of interest. Monetary equilibrium occurs at the rate of interest where LP and M_s intersect.



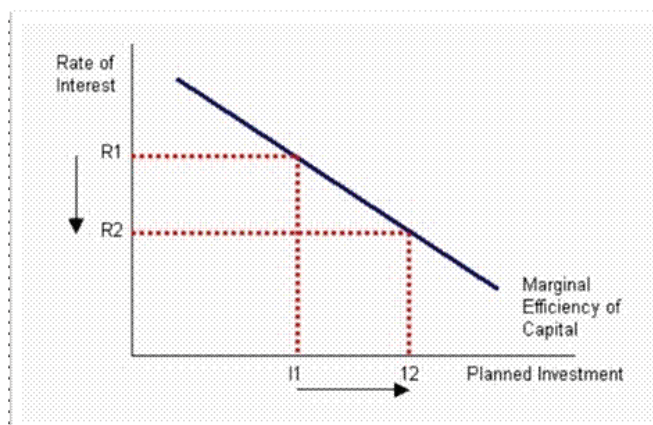
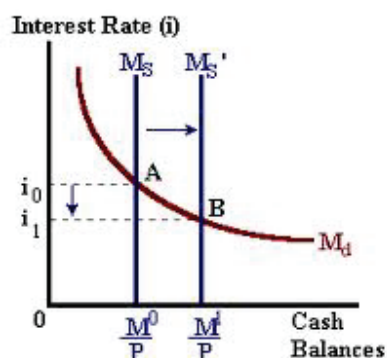
Keynesian money transmission mechanism:

The mechanism by which changes in the demand for and supply of money affect aggregate demand is called the transmission mechanism. It operates in three states; the first is the link between monetary equilibrium and the rate of interest, the second is the link between the interest rate and investment expenditure, and the third is the link between investment expenditure and aggregate demand.

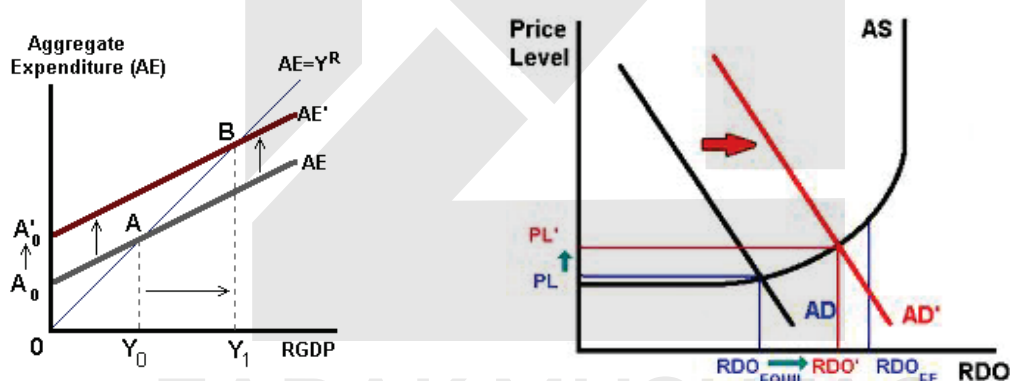
An increase in the supply of money, with an unchanged money demand function, will give rise to an excess supply of money at the original interest rate. An excess supply of money will cause the interest rate to fall. Investment, which includes expenditures on inventory accumulation, residential construction, and business fixed investment, responds to changes in their real rate of interest. Other things being equal, a decrease in the real interest makes borrowing cheaper and generates new investment expenditure.

There is no such thing as luck, believe in hard work

Real national income (Y) and the price level



The first two links in the transmission is shown in the above figures i.e. a change in the money supply causes the rate of interest to change in the opposite direction and a change in the interest rate causes the level of investment expenditure to change in the opposite direction. Thus, changes in the money supply cause investment expenditure to change in the same direction i.e. and increase in the money supply leads to a fall in the interest rate and an increase in investment expenditure and a decrease in the money supply leads to a rise in the interest rate and a decrease in investment expenditure.



The third link is between the investment expenditure and the aggregate demand. A change in the money supply, by causing a change in desired investment expenditure and hence a shift in the aggregate expenditure curve. An increase in the money supply causes an increase in investment expenditure and therefore an increase in aggregate demand. The transmission mechanism connects monetary forces and real expenditure flows. It works from a change in the demand for or supply of money to a change in bond prices and interest rate, to changes in investment expenditures and to shift in AD curve.

The loanable funds theory: Loanable funds represent the money in commercial banks and lending institutions that is available to lend out to firms and households to finance expenditures (investment or consumption). The Y-axis represents the *real interest rate*; the loanable funds

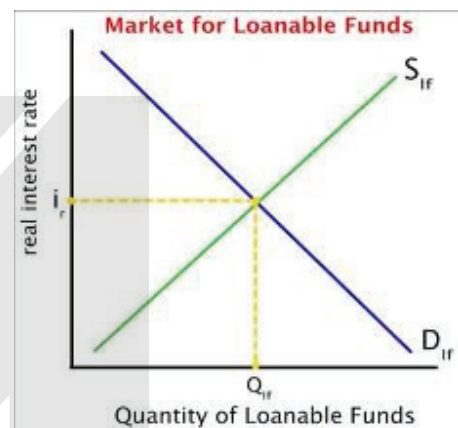
There is no such thing as luck, believe in hard work

Real national income (Y) and the price level

market therefore recognizes the relationships between real returns on savings and real price of borrowing with the public's willingness to save and borrow.

Since an increase in the real interest rate makes households and firms want to place more money in the bank (and more money in the bank means more money to loan out), there is a direct relationship between real interest rate and Supply of Loanable Funds. On the other hand, since at lower real interest rates households and firms will be less inclined to save and more inclined to borrow and spend.

The demand for loanable funds reflects an inverse relationship. At higher interest rates, households prefer to delay their spending and put their money in savings, since the opportunity cost of spending now rises with the real interest rate. The supply of funds primarily depends on saving activity of people in the economy which in turn depends on the rate of interest. Bank gives interest reward to the savers and provide or supply these fund to the borrowers at a higher rate of interest, the difference of these rate of interest is the income of the bank called the spread.



ZARAK MUSHTAQ

There is no such thing as luck, believe in hard work